

Colliding Wind Apep with a Supernova Event

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Project Description

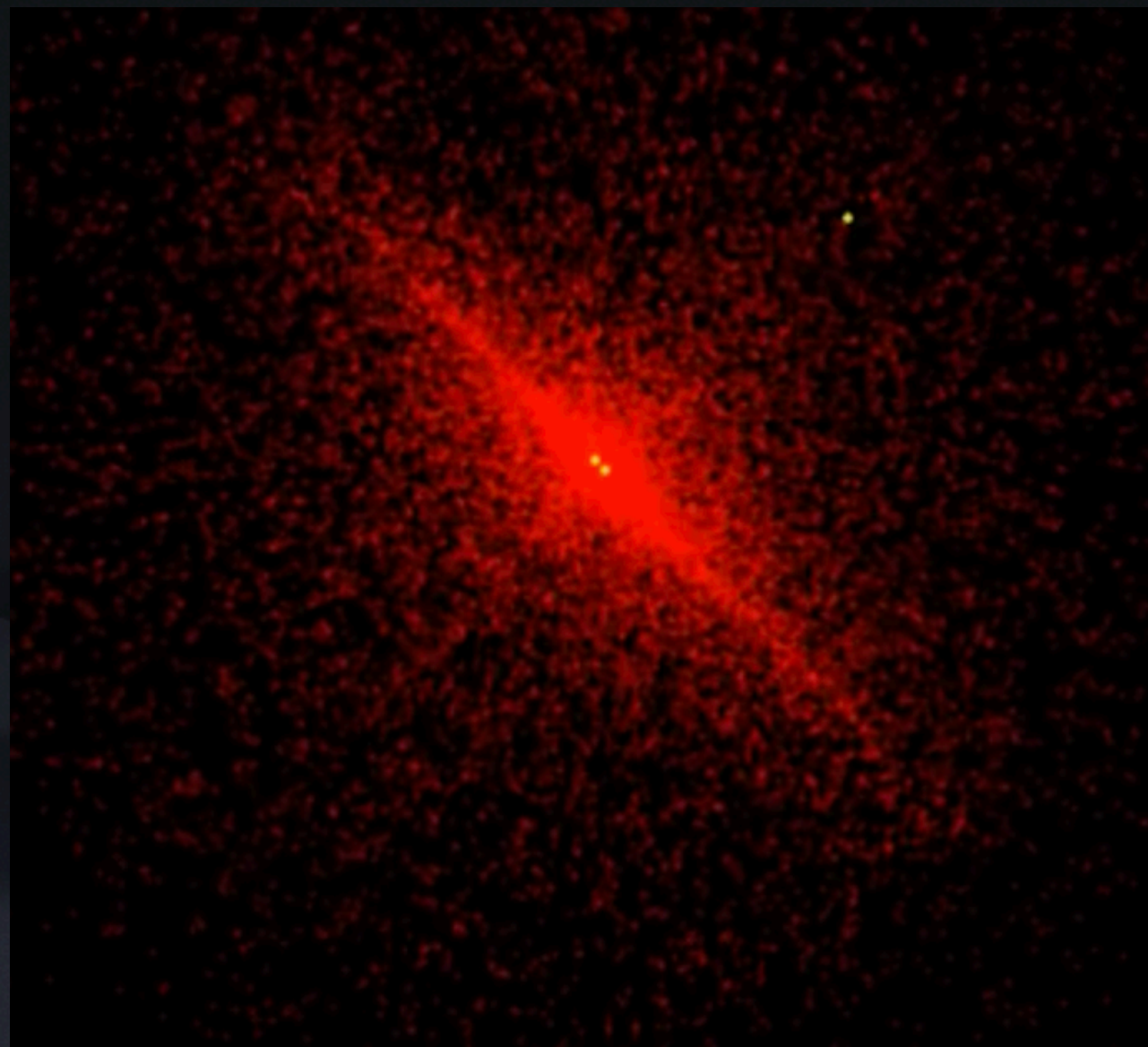
The Apep System

- Tried to reproduce the Apep System- a complex triple stellar system located in the Norma constellation consisting of a central binary system of two Wolf-Rayet (WR) stars of subtypes WC8 and WN4-6b, and an O8 Iaf supergiant positioned approximately 1700 AU to the north.
- This system represents one of the strongest cases of a classical WR+WR binary system in our galaxy.
- Made the WN4-6b go supernova really early to see the change in morphology.

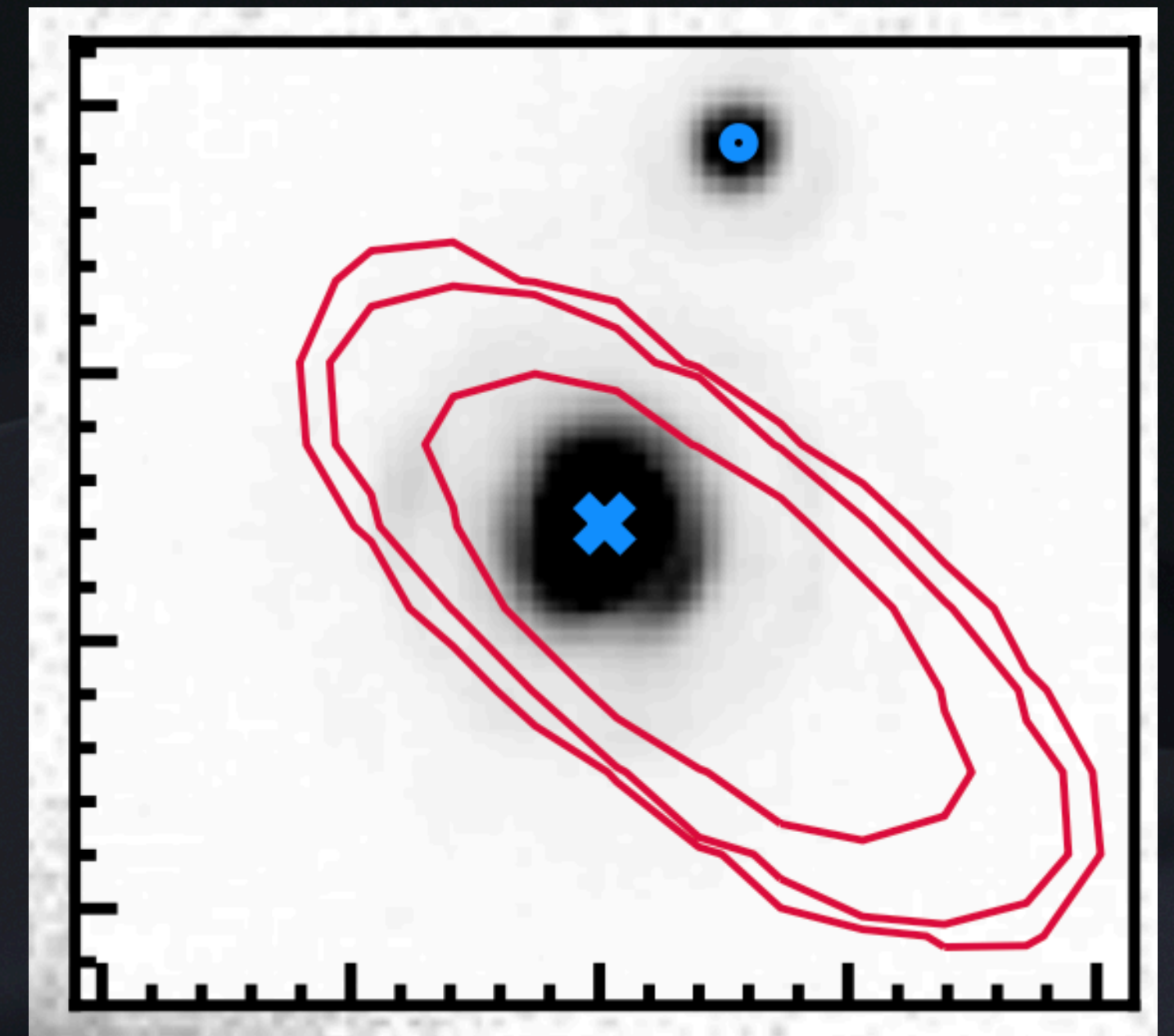
More about Apep

What we saw in the literature (and interview with Joe)

- The Central Engine and the northern companion
- Really zoomed in to resolve the central engine



Our Result



Credit: J. R. Callingham et al.

Initializing apep

As realistic as possible

- WN46b Wolf-Rayet
 - 11 Solar masses, 6 Solar radii
 - $4e-5$ Solar mass/yr mass loss
- WC8 Wolf-Rayet
 - 18 Solar masses, 6.3 Solar radii
 - $3e-5$ Solar mass/yr mass loss
- 'Central engine' = WR-binary → separation of 67 AU, period of 125 yr, eccentricity 0.7
- O8 Iaf Supergiant
 - 40 Solar masses, 35.4 Solar radii
 - 1700 AU from the Central Engine, no stellar winds

Initializing apep

Stellar winds

- WN46b Wolf-Rayet
 - Terminal wind velocity = 3500 km/s
 - Wind temperature is 30% of the star temperature
- WC8 Wolf-Rayet
 - Terminal wind velocity = 2100 km/s
 - Rotational velocity of ~500 km/s around its own axis

Methodology

Binary System Dynamics: N-body Code- ph4

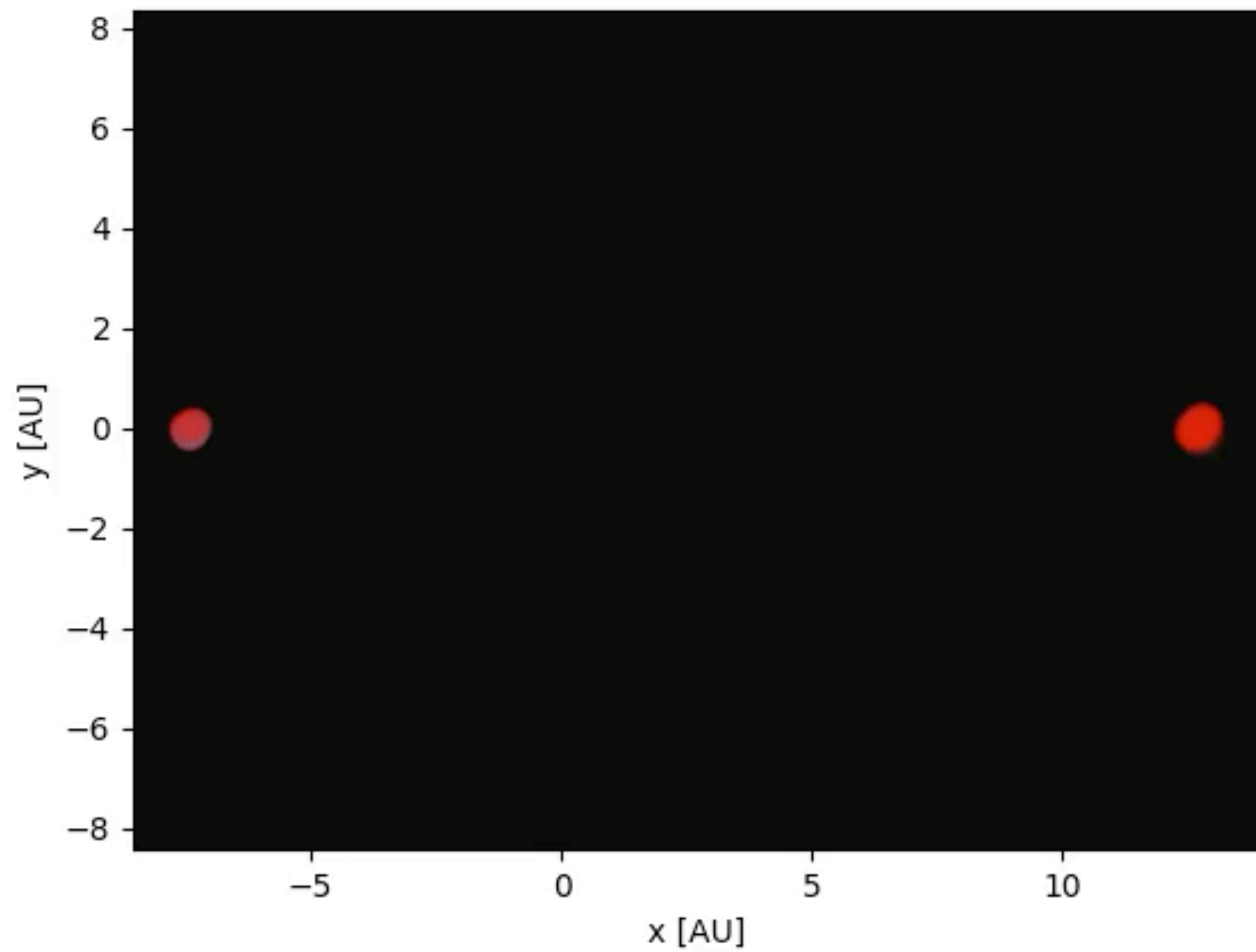
- For the gravitational interactions between the stars in the Apep system. Mainly for the orbital evolution of the stars.

Hydrodynamics: Fi

- Used to simulate the wind-wind collision between the WR stars and later interactions with the supernova's winds. Modelling the shocks and any other hydrodynamical interactions.

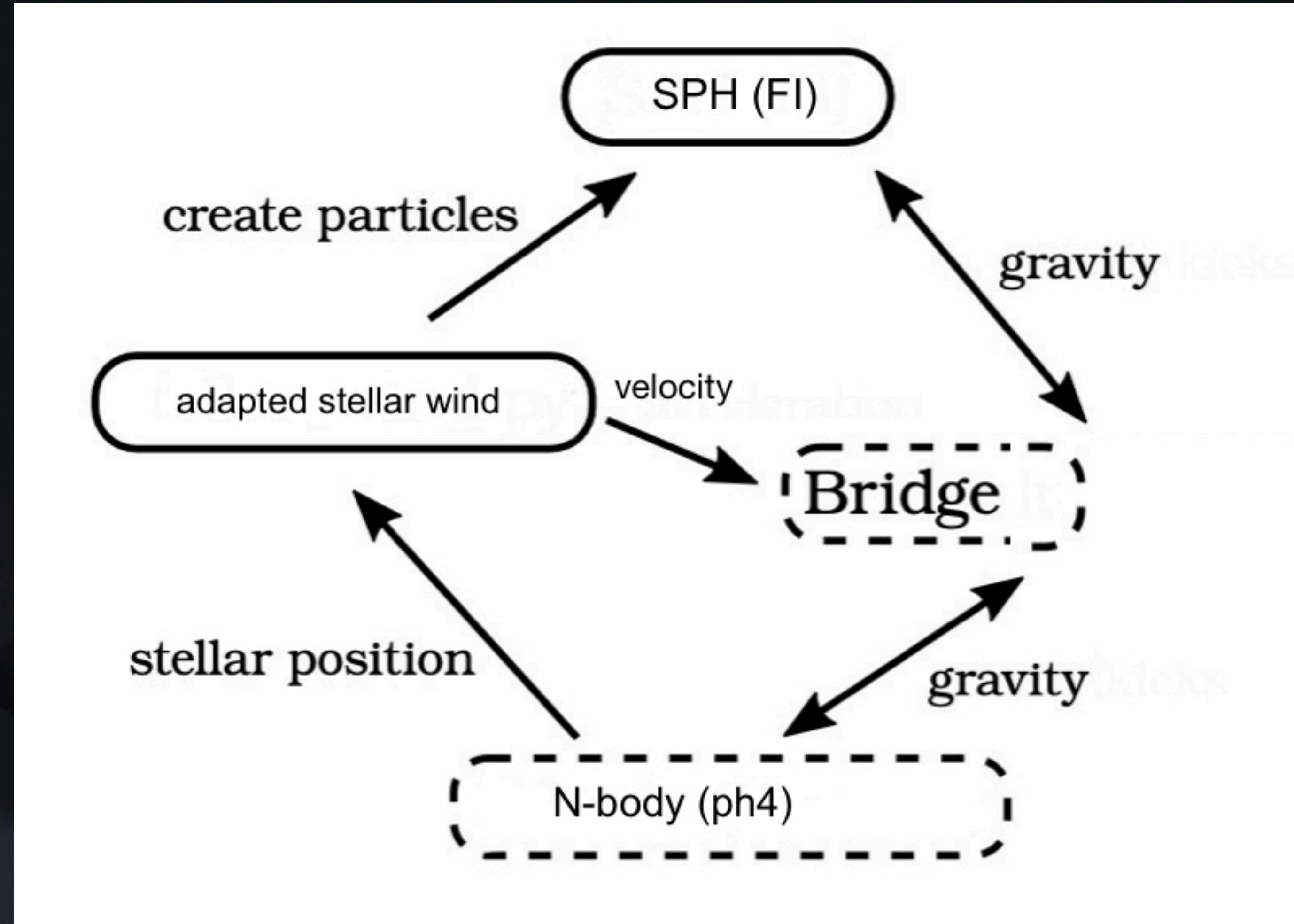
Stellar Evolution: MESA

- Initially thought of evolving the stars individually using SeBa, but saw a little influence on the morphology. MESA is used only once to generate the supernova.



Bridge

Making Gravity and Hydro work together



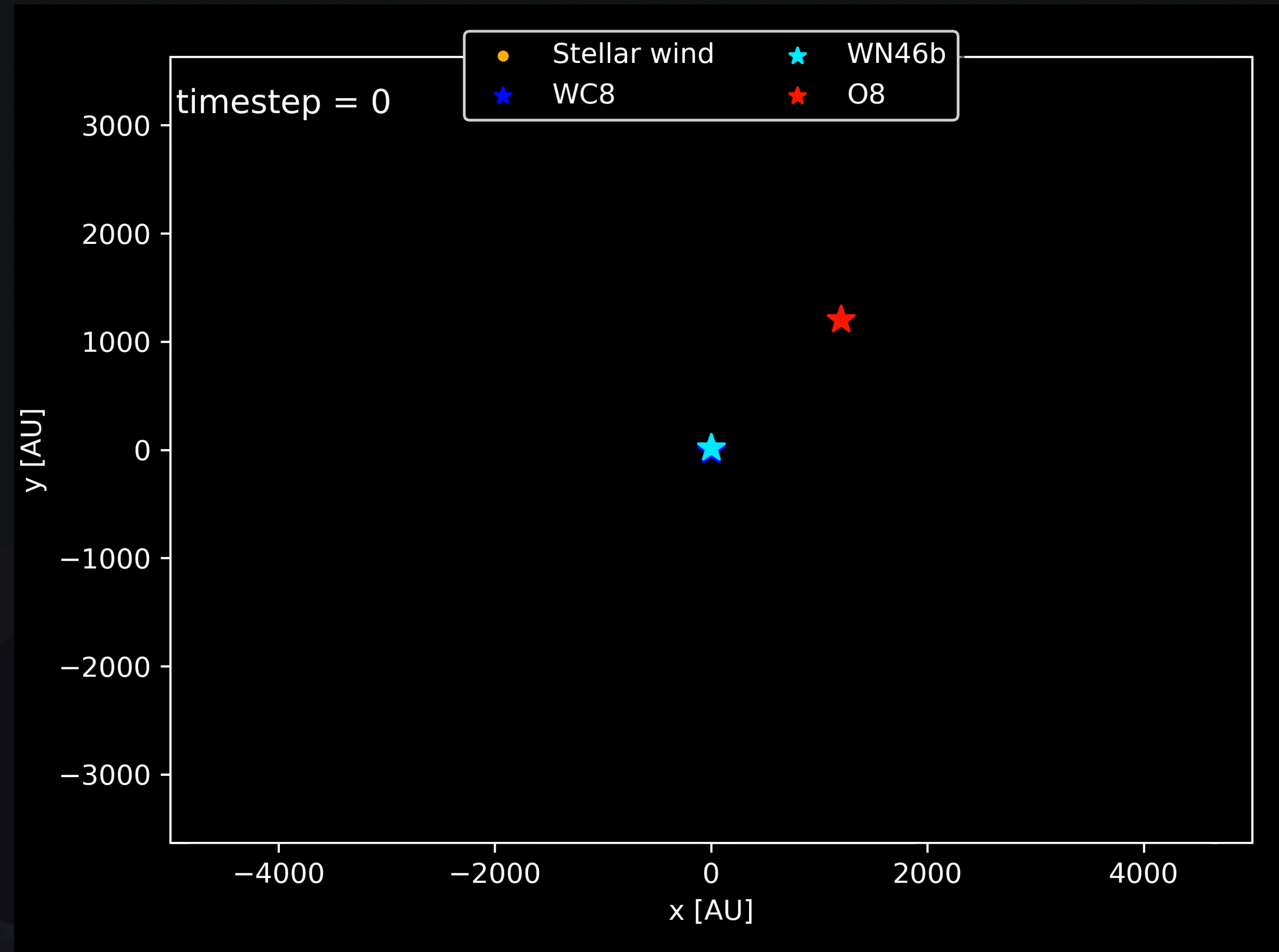
Results

Our First attempt:

- We made the wind using an adaptation of AMUSE's stellar wind module

Results

Our First attempt:

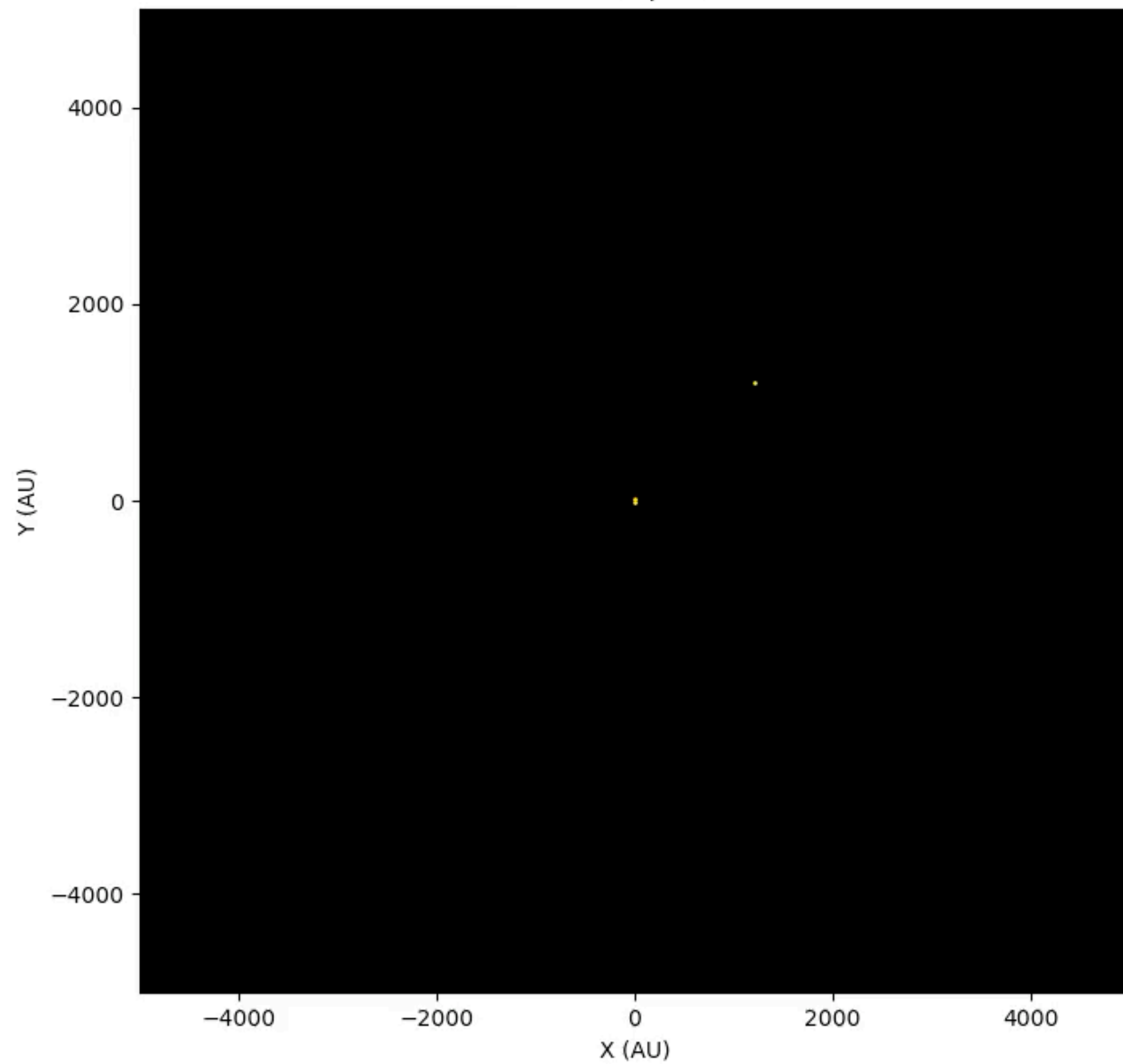


Results

Second attempt: adding rotation

- Adding tangential velocity of ~ 500 km/s to the SPH particles coming off WC8 star

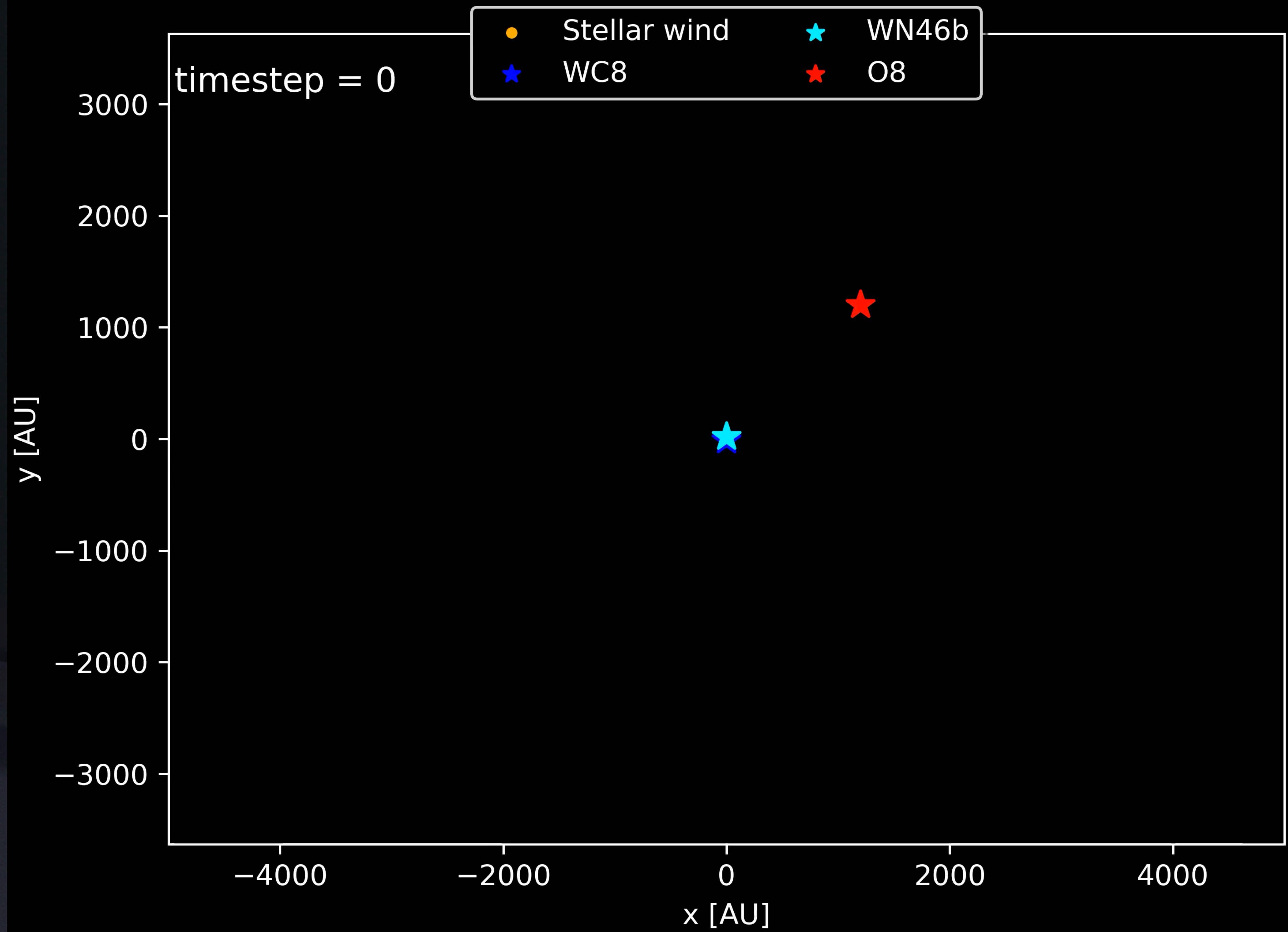
timestep = 0



Results

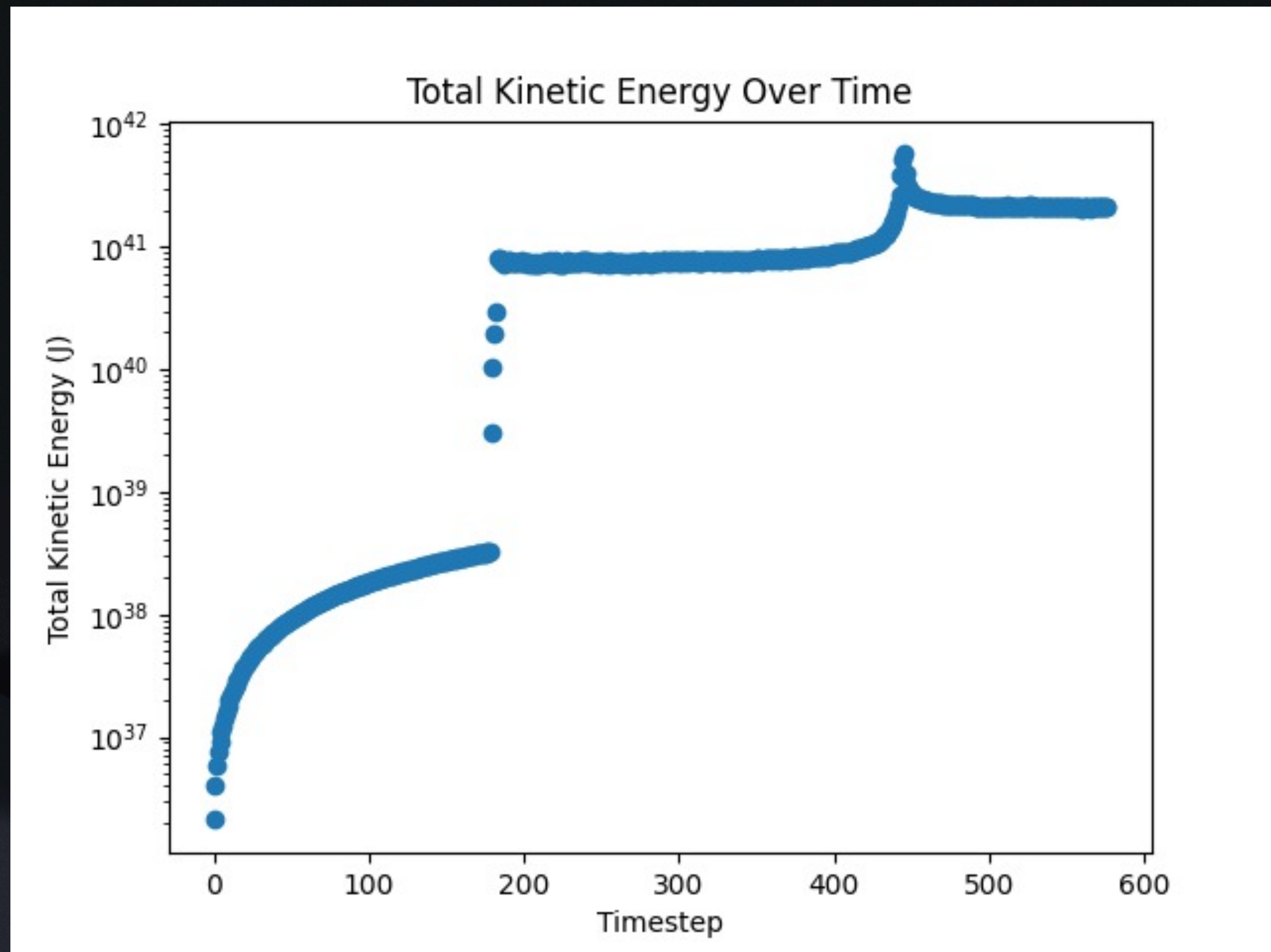
What would happen if WN46b goes supernova?

- This will happen sometime in the future, it will be the first star of the system to go supernova
- We expect influence on the morphology of the stellar wind



Results

Supernova



Future work

- See which supernova energy is most likely to significantly affect the morphology of the stellar wind
- Run the simulation longer to find out what will happen to Apep in the future (and at different viewing angles)

Thank you!

Any questions?